

# DAC Enode Intelligence Watcher - Custom All Time Report

Generated at UTC: 2026-09-19T18:36:50.690801+00:00

Report range: All Time

Project: DAC Enode Intelligence Watcher

This custom report is generated from local watcher JSON outputs.

Important note: this report is observation-based. It does not make official DAC ownership claims and should not be treated as a definitive decentralization measurement.

## 1. Report Scope

| Field                         | Value                            |
|-------------------------------|----------------------------------|
| Range                         | All Time                         |
| Observation count             | 198                              |
| First observed source time    | May 15, 2026 (00:00 CEST)        |
| Last observed source time     | 2026-09-19 05:00 CEST            |
| Latest watcher checked_at_utc | 2026-09-19T03:29:30.338786+00:00 |
| Latest source generated time  | 2026-09-19 05:00 CEST            |

## 2. Enode Movement Summary

| Metric                     | Value |
|----------------------------|-------|
| Minimum enode count        | 2     |
| Maximum enode count        | 16    |
| Average enode count        | 10.73 |
| Total added observations   | 183   |
| Total removed observations | 161   |

| Metric                | Value |
|-----------------------|-------|
| Target ports observed | 28657 |

### 3. Observation Source Coverage

| Phase             | Observations |
|-------------------|--------------|
| automated_watcher | 184          |
| manual_backfill   | 14           |

### 4. Anomaly Summary

| Field                    | Value                                                                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------------|
| Selected anomaly signals | 20                                                                                                           |
| Global anomaly summary   | 20 anomaly signals were detected across 198 observations. The highest observed anomaly severity is CRITICAL. |
| Global highest severity  | CRITICAL                                                                                                     |
| Recommended action       | Use these anomaly events as candidates for deeper manual review and future technical reporting.              |

| Severity | Signals in selected range |
|----------|---------------------------|
| HIGH     | 12                        |
| CRITICAL | 4                         |
| MEDIUM   | 4                         |

| Anomaly Type                 | Signals in selected range |
|------------------------------|---------------------------|
| HIGH_REMOVAL_EVENT           | 5                         |
| AGGRESSIVE_ROTATION          | 5                         |
| WATCHER_HIGH_SEVERITY_SIGNAL | 3                         |
| SHARP_ENODE_COUNT_DROP       | 2                         |
| LOW_CONTINUITY_RATIO         | 2                         |

| Anomaly Type            | Signals in selected range |
|-------------------------|---------------------------|
| MODERATE_ROTATION_SPIKE | 2                         |
| LARGE_ENODE_COUNT_DROP  | 1                         |

## 5. Provider / ASN Concentration Context

| Field                       | Value                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overall concentration label | MODERATE                                                                                                                                                                                                                                                                                                                                   |
| Headline                    | Observed infrastructure shows moderate concentration under the current heuristic model.                                                                                                                                                                                                                                                    |
| Key observation             | Top live ASN is AS51167 with 16 unique IPs (35.56%).                                                                                                                                                                                                                                                                                       |
| Country observation         | Top live ASN country code is DE with 19 unique IPs (42.22%).                                                                                                                                                                                                                                                                               |
| Interpretation              | Top live ASN controls at least 35% of observed unique IPs.                                                                                                                                                                                                                                                                                 |
| Disclaimer                  | Provider concentration and decentralization risk summary is an observation-based heuristic. It is based on currently available watcher data, live ASN enrichment, static provider hints, and DAC Infrastructure Signal labels. It should not be treated as an official DAC classification or as a definitive decentralization measurement. |

| Dimension                 | Top Name            | Top Count | Top % | Label    |
|---------------------------|---------------------|-----------|-------|----------|
| Live ASN                  | AS51167             | 16        | 35.56 | MODERATE |
| Live Country              | DE                  | 19        | 42.22 | MODERATE |
| DAC Infrastructure Signal | Unknown / No Signal | 15        | 33.33 | LOW      |

## 6. Observation Timeline

| # | Phase           | Status         | Source Time               | Current | Added | Removed | Unchanged | Port  |
|---|-----------------|----------------|---------------------------|---------|-------|---------|-----------|-------|
| 1 | manual_backfill | manual_initial | May 15, 2026 (00:00 CEST) | 11      | 11    | 0       | 0         | 28657 |
| 2 | manual_backfill | manual_changed | May 16, 2026 (20:00 CEST) | 14      | 4     | 1       | 10        | 28657 |
| 3 | manual_backfill | manual_changed | May 17, 2026 (16:00 CEST) | 13      | 1     | 2       | 12        | 28657 |
| 4 | manual_backfill | manual_changed | May 18, 2026 (08:00 CEST) | 13      | 1     | 1       | 12        | 28657 |

| #  | Phase             | Status           | Source Time               | Current | Added | Removed | Unchanged | Port  |
|----|-------------------|------------------|---------------------------|---------|-------|---------|-----------|-------|
| 5  | manual_backfill   | manual_changed   | May 19, 2026 (16:00 CEST) | 7       | 1     | 7       | 6         | 28657 |
| 6  | manual_backfill   | manual_changed   | May 20, 2026 (20:00 CEST) | 12      | 5     | 0       | 7         | 28657 |
| 7  | manual_backfill   | manual_changed   | May 22, 2026 (00:00 CEST) | 12      | 1     | 1       | 11        | 28657 |
| 8  | manual_backfill   | manual_changed   | May 23, 2026 (12:00 CEST) | 12      | 4     | 4       | 8         | 28657 |
| 9  | manual_backfill   | manual_changed   | May 24, 2026 (04:00 CEST) | 9       | 0     | 3       | 9         | 28657 |
| 10 | manual_backfill   | manual_changed   | May 25, 2026 (04:00 CEST) | 12      | 3     | 0       | 9         | 28657 |
| 11 | manual_backfill   | manual_changed   | May 28, 2026 (08:00 CEST) | 14      | 5     | 3       | 9         | 28657 |
| 12 | manual_backfill   | manual_unchanged | May 29, 2026 (04:00 CEST) | 14      | 0     | 0       | 14        | 28657 |
| 13 | manual_backfill   | manual_changed   | May 30, 2026 (12:00 CEST) | 14      | 1     | 1       | 13        | 28657 |
| 14 | manual_backfill   | manual_changed   | May 31, 2026 (12:00 CEST) | 15      | 2     | 1       | 13        | 28657 |
| 15 | automated_watcher | initial          | Jun 1, 2026 (04:00 CEST)  | 12      | 12    | 0       | 0         | 28657 |
| 16 | automated_watcher | changed          | Jun 1, 2026 (08:00 CEST)  | 13      | 1     | 0       | 12        | 28657 |
| 17 | automated_watcher | changed          | Jun 1, 2026 (12:00 CEST)  | 9       | 0     | 4       | 9         | 28657 |
| 18 | automated_watcher | changed          | Jun 1, 2026 (22:00 CEST)  | 10      | 1     | 0       | 9         | 28657 |
| 19 | automated_watcher | changed          | Jun 2, 2026 (06:00 CEST)  | 12      | 2     | 0       | 10        | 28657 |
| 20 | automated_watcher | changed          | Jun 2, 2026 (12:00 CEST)  | 12      | 1     | 1       | 11        | 28657 |
| 21 | automated_watcher | changed          | Jun 2, 2026 (22:00 CEST)  | 14      | 2     | 0       | 12        | 28657 |
| 22 | automated_watcher | changed          | Jun 3, 2026 (12:00 CEST)  | 10      | 0     | 4       | 10        | 28657 |
| 23 | automated_watcher | changed          | Jun 3, 2026 (14:00 CEST)  | 8       | 0     | 2       | 8         | 28657 |

| #  | Phase             | Status  | Source Time              | Current | Added | Removed | Unchanged | Port  |
|----|-------------------|---------|--------------------------|---------|-------|---------|-----------|-------|
| 24 | automated_watcher | changed | Jun 3, 2026 (22:00 CEST) | 10      | 2     | 0       | 8         | 28657 |
| 25 | automated_watcher | changed | Jun 4, 2026 (00:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 26 | automated_watcher | changed | Jun 4, 2026 (02:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 27 | automated_watcher | changed | Jun 4, 2026 (12:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 28 | automated_watcher | changed | Jun 4, 2026 (18:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 29 | automated_watcher | changed | Jun 6, 2026 (08:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 30 | automated_watcher | changed | Jun 6, 2026 (10:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 31 | automated_watcher | changed | Jun 6, 2026 (14:00 CEST) | 13      | 1     | 0       | 12        | 28657 |
| 32 | automated_watcher | changed | Jun 6, 2026 (16:00 CEST) | 14      | 1     | 0       | 13        | 28657 |
| 33 | automated_watcher | changed | Jun 6, 2026 (20:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 34 | automated_watcher | changed | Jun 6, 2026 (22:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 35 | automated_watcher | changed | Jun 7, 2026 (02:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 36 | automated_watcher | changed | Jun 7, 2026 (08:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 37 | automated_watcher | changed | Jun 7, 2026 (10:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 38 | automated_watcher | changed | Jun 7, 2026 (14:00 CEST) | 15      | 1     | 1       | 14        | 28657 |
| 39 | automated_watcher | changed | Jun 7, 2026 (20:00 CEST) | 16      | 1     | 0       | 15        | 28657 |
| 40 | automated_watcher | changed | Jun 8, 2026 (06:00 CEST) | 15      | 0     | 1       | 15        | 28657 |
| 41 | automated_watcher | changed | Jun 8, 2026 (20:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 42 | automated_watcher | changed | Jun 8, 2026 (22:00 CEST) | 15      | 1     | 0       | 14        | 28657 |

| #  | Phase             | Status  | Source Time               | Current | Added | Removed | Unchanged | Port  |
|----|-------------------|---------|---------------------------|---------|-------|---------|-----------|-------|
| 43 | automated_watcher | changed | Jun 9, 2026 (00:00 CEST)  | 14      | 0     | 1       | 14        | 28657 |
| 44 | automated_watcher | changed | Jun 9, 2026 (02:00 CEST)  | 12      | 0     | 2       | 12        | 28657 |
| 45 | automated_watcher | changed | Jun 9, 2026 (06:00 CEST)  | 11      | 0     | 1       | 11        | 28657 |
| 46 | automated_watcher | changed | Jun 9, 2026 (10:00 CEST)  | 10      | 0     | 1       | 10        | 28657 |
| 47 | automated_watcher | changed | Jun 9, 2026 (12:00 CEST)  | 11      | 1     | 0       | 10        | 28657 |
| 48 | automated_watcher | changed | Jun 9, 2026 (16:00 CEST)  | 10      | 0     | 1       | 10        | 28657 |
| 49 | automated_watcher | changed | Jun 9, 2026 (20:00 CEST)  | 11      | 1     | 0       | 10        | 28657 |
| 50 | automated_watcher | changed | Jun 9, 2026 (22:00 CEST)  | 12      | 1     | 0       | 11        | 28657 |
| 51 | automated_watcher | changed | Jun 10, 2026 (06:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 52 | automated_watcher | changed | Jun 10, 2026 (10:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 53 | automated_watcher | changed | Jun 10, 2026 (14:00 CEST) | 13      | 1     | 0       | 12        | 28657 |
| 54 | automated_watcher | changed | Jun 10, 2026 (18:00 CEST) | 14      | 1     | 0       | 13        | 28657 |
| 55 | automated_watcher | changed | Jun 11, 2026 (00:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 56 | automated_watcher | changed | Jun 11, 2026 (02:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 57 | automated_watcher | changed | Jun 11, 2026 (08:00 CEST) | 14      | 1     | 1       | 13        | 28657 |
| 58 | automated_watcher | changed | Jun 11, 2026 (12:00 CEST) | 13      | 0     | 1       | 13        | 28657 |
| 59 | automated_watcher | changed | Jun 11, 2026 (20:00 CEST) | 14      | 1     | 0       | 13        | 28657 |
| 60 | automated_watcher | changed | Jun 11, 2026 (22:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 61 | automated_watcher | changed | Jun 12, 2026 (16:00 CEST) | 14      | 0     | 1       | 14        | 28657 |

| #  | Phase             | Status  | Source Time               | Current | Added | Removed | Unchanged | Port  |
|----|-------------------|---------|---------------------------|---------|-------|---------|-----------|-------|
| 62 | automated_watcher | changed | Jun 12, 2026 (20:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 63 | automated_watcher | changed | Jun 12, 2026 (22:00 CEST) | 16      | 1     | 0       | 15        | 28657 |
| 64 | automated_watcher | changed | Jun 13, 2026 (00:00 CEST) | 15      | 0     | 1       | 15        | 28657 |
| 65 | automated_watcher | changed | Jun 13, 2026 (04:00 CEST) | 16      | 1     | 0       | 15        | 28657 |
| 66 | automated_watcher | changed | Jun 13, 2026 (08:00 CEST) | 15      | 0     | 1       | 15        | 28657 |
| 67 | automated_watcher | changed | Jun 13, 2026 (22:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 68 | automated_watcher | changed | Jun 14, 2026 (02:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 69 | automated_watcher | changed | Jun 14, 2026 (12:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 70 | automated_watcher | changed | Jun 14, 2026 (14:00 CEST) | 16      | 2     | 0       | 14        | 28657 |
| 71 | automated_watcher | changed | Jun 14, 2026 (18:00 CEST) | 14      | 0     | 2       | 14        | 28657 |
| 72 | automated_watcher | changed | Jun 14, 2026 (20:00 CEST) | 14      | 1     | 1       | 13        | 28657 |
| 73 | automated_watcher | changed | Jun 15, 2026 (08:00 CEST) | 13      | 0     | 1       | 13        | 28657 |
| 74 | automated_watcher | changed | Jun 15, 2026 (20:00 CEST) | 11      | 0     | 2       | 11        | 28657 |
| 75 | automated_watcher | changed | Jun 16, 2026 (04:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 76 | automated_watcher | changed | Jun 16, 2026 (14:00 CEST) | 11      | 2     | 1       | 9         | 28657 |
| 77 | automated_watcher | changed | Jun 16, 2026 (18:00 CEST) | 8       | 0     | 3       | 8         | 28657 |
| 78 | automated_watcher | changed | Jun 16, 2026 (22:00 CEST) | 9       | 1     | 0       | 8         | 28657 |
| 79 | automated_watcher | changed | Jun 17, 2026 (06:00 CEST) | 10      | 1     | 0       | 9         | 28657 |
| 80 | automated_watcher | changed | Jun 17, 2026 (12:00 CEST) | 2       | 0     | 8       | 2         | 28657 |

| #  | Phase             | Status  | Source Time               | Current | Added | Removed | Unchanged | Port  |
|----|-------------------|---------|---------------------------|---------|-------|---------|-----------|-------|
| 81 | automated_watcher | changed | Jun 17, 2026 (16:00 CEST) | 6       | 5     | 1       | 1         | 28657 |
| 82 | automated_watcher | changed | Jun 17, 2026 (18:00 CEST) | 7       | 1     | 0       | 6         | 28657 |
| 83 | automated_watcher | changed | Jun 17, 2026 (20:00 CEST) | 7       | 1     | 1       | 6         | 28657 |
| 84 | automated_watcher | changed | Jun 18, 2026 (00:00 CEST) | 8       | 1     | 0       | 7         | 28657 |
| 85 | automated_watcher | changed | Jun 18, 2026 (08:00 CEST) | 7       | 0     | 1       | 7         | 28657 |
| 86 | automated_watcher | changed | Jun 18, 2026 (13:00 CEST) | 10      | 3     | 0       | 7         | 28657 |
| 87 | automated_watcher | changed | Jun 18, 2026 (16:00 CEST) | 9       | 1     | 2       | 8         | 28657 |
| 88 | automated_watcher | changed | Jun 18, 2026 (19:00 CEST) | 8       | 0     | 1       | 8         | 28657 |
| 89 | automated_watcher | changed | Jun 19, 2026 (00:00 CEST) | 9       | 1     | 0       | 8         | 28657 |
| 90 | automated_watcher | changed | Jun 19, 2026 (07:00 CEST) | 8       | 0     | 1       | 8         | 28657 |
| 91 | automated_watcher | changed | Jun 19, 2026 (11:00 CEST) | 6       | 1     | 3       | 5         | 28657 |
| 92 | automated_watcher | changed | Jun 19, 2026 (15:00 CEST) | 9       | 3     | 0       | 6         | 28657 |
| 93 | automated_watcher | changed | Jun 19, 2026 (18:00 CEST) | 9       | 1     | 1       | 8         | 28657 |
| 94 | automated_watcher | changed | Jun 19, 2026 (20:00 CEST) | 8       | 0     | 1       | 8         | 28657 |
| 95 | automated_watcher | changed | Jun 19, 2026 (22:00 CEST) | 8       | 1     | 1       | 7         | 28657 |
| 96 | automated_watcher | changed | Jun 20, 2026 (00:00 CEST) | 8       | 1     | 1       | 7         | 28657 |
| 97 | automated_watcher | changed | Jun 20, 2026 (05:00 CEST) | 10      | 2     | 0       | 8         | 28657 |
| 98 | automated_watcher | changed | Jun 20, 2026 (16:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 99 | automated_watcher | changed | Jun 20, 2026 (21:00 CEST) | 11      | 1     | 1       | 10        | 28657 |

| #   | Phase             | Status  | Source Time               | Current | Added | Removed | Unchanged | Port  |
|-----|-------------------|---------|---------------------------|---------|-------|---------|-----------|-------|
| 100 | automated_watcher | changed | Jun 21, 2026 (00:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 101 | automated_watcher | changed | Jun 21, 2026 (01:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 102 | automated_watcher | changed | Jun 21, 2026 (09:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 103 | automated_watcher | changed | Jun 21, 2026 (15:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 104 | automated_watcher | changed | Jun 21, 2026 (22:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 105 | automated_watcher | changed | Jun 21, 2026 (23:00 CEST) | 10      | 1     | 1       | 9         | 28657 |
| 106 | automated_watcher | changed | Jun 22, 2026 (09:00 CEST) | 8       | 0     | 2       | 8         | 28657 |
| 107 | automated_watcher | changed | Jun 22, 2026 (15:00 CEST) | 6       | 0     | 2       | 6         | 28657 |
| 108 | automated_watcher | changed | Jun 22, 2026 (19:00 CEST) | 9       | 3     | 0       | 6         | 28657 |
| 109 | automated_watcher | changed | Jun 22, 2026 (22:00 CEST) | 10      | 2     | 1       | 8         | 28657 |
| 110 | automated_watcher | changed | Jun 23, 2026 (00:00 CEST) | 9       | 0     | 1       | 9         | 28657 |
| 111 | automated_watcher | changed | Jun 23, 2026 (02:00 CEST) | 10      | 1     | 0       | 9         | 28657 |
| 112 | automated_watcher | changed | Jun 23, 2026 (06:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 113 | automated_watcher | changed | Jun 23, 2026 (10:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 114 | automated_watcher | changed | Jun 23, 2026 (16:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 115 | automated_watcher | changed | Jun 23, 2026 (18:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 116 | automated_watcher | changed | Jun 24, 2026 (01:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 117 | automated_watcher | changed | Jun 24, 2026 (03:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 118 | automated_watcher | changed | Jun 24, 2026 (07:00 CEST) | 13      | 1     | 0       | 12        | 28657 |

| #   | Phase             | Status  | Source Time               | Current | Added | Removed | Unchanged | Port  |
|-----|-------------------|---------|---------------------------|---------|-------|---------|-----------|-------|
| 119 | automated_watcher | changed | Jun 24, 2026 (14:00 CEST) | 12      | 0     | 1       | 12        | 28657 |
| 120 | automated_watcher | changed | Jun 24, 2026 (22:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 121 | automated_watcher | changed | Jun 25, 2026 (00:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 122 | automated_watcher | changed | Jun 25, 2026 (21:00 CEST) | 7       | 0     | 3       | 7         | 28657 |
| 123 | automated_watcher | changed | Jun 26, 2026 (02:00 CEST) | 10      | 3     | 0       | 7         | 28657 |
| 124 | automated_watcher | changed | Jun 26, 2026 (10:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 125 | automated_watcher | changed | Jun 26, 2026 (21:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 126 | automated_watcher | changed | Jun 27, 2026 (05:00 CEST) | 10      | 0     | 2       | 10        | 28657 |
| 127 | automated_watcher | changed | Jun 27, 2026 (08:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 128 | automated_watcher | changed | Jun 27, 2026 (11:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 129 | automated_watcher | changed | Jun 27, 2026 (16:00 CEST) | 13      | 1     | 0       | 12        | 28657 |
| 130 | automated_watcher | changed | Jun 28, 2026 (10:00 CEST) | 12      | 0     | 1       | 12        | 28657 |
| 131 | automated_watcher | changed | Jun 28, 2026 (12:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 132 | automated_watcher | changed | Jun 28, 2026 (14:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 133 | automated_watcher | changed | Jun 28, 2026 (19:00 CEST) | 13      | 1     | 0       | 12        | 28657 |
| 134 | automated_watcher | changed | Jun 30, 2026 (18:00 CEST) | 14      | 1     | 0       | 13        | 28657 |
| 135 | automated_watcher | changed | Jun 30, 2026 (22:00 CEST) | 13      | 0     | 1       | 13        | 28657 |
| 136 | automated_watcher | changed | Jul 1, 2026 (01:00 CEST)  | 12      | 0     | 1       | 12        | 28657 |
| 137 | automated_watcher | changed | Jul 1, 2026 (12:00 CEST)  | 13      | 1     | 0       | 12        | 28657 |
| 138 | automated_watcher | changed | Jul 1, 2026 (22:00 CEST)  | 14      | 1     | 0       | 13        | 28657 |

| #   | Phase             | Status  | Source Time               | Current | Added | Removed | Unchanged | Port  |
|-----|-------------------|---------|---------------------------|---------|-------|---------|-----------|-------|
| 139 | automated_watcher | changed | Jul 2, 2026 (13:00 CEST)  | 15      | 1     | 0       | 14        | 28657 |
| 140 | automated_watcher | changed | Jul 3, 2026 (19:00 CEST)  | 14      | 0     | 1       | 14        | 28657 |
| 141 | automated_watcher | changed | Jul 4, 2026 (07:00 CEST)  | 13      | 0     | 1       | 13        | 28657 |
| 142 | automated_watcher | changed | Jul 4, 2026 (18:00 CEST)  | 12      | 0     | 1       | 12        | 28657 |
| 143 | automated_watcher | changed | Jul 8, 2026 (11:00 CEST)  | 13      | 1     | 0       | 12        | 28657 |
| 144 | automated_watcher | changed | Jul 13, 2026 (08:00 CEST) | 11      | 0     | 2       | 11        | 28657 |
| 145 | automated_watcher | changed | Jul 14, 2026 (16:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 146 | automated_watcher | changed | Jul 14, 2026 (17:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 147 | automated_watcher | changed | Jul 15, 2026 (16:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 148 | automated_watcher | changed | Jul 20, 2026 (15:00 CEST) | 13      | 1     | 0       | 12        | 28657 |
| 149 | automated_watcher | changed | Jul 23, 2026 (03:00 CEST) | 14      | 1     | 0       | 13        | 28657 |
| 150 | automated_watcher | changed | Jul 23, 2026 (13:00 CEST) | 13      | 0     | 1       | 13        | 28657 |
| 151 | automated_watcher | changed | 2026-07-24 06:00 CEST     | 14      | 1     | 0       | 13        | 28657 |
| 152 | automated_watcher | changed | 2026-07-28 09:00 CEST     | 12      | 0     | 2       | 12        | 28657 |
| 153 | automated_watcher | changed | 2026-07-29 13:00 CEST     | 13      | 1     | 0       | 12        | 28657 |
| 154 | automated_watcher | changed | 2026-07-31 11:00 CEST     | 14      | 1     | 0       | 13        | 28657 |
| 155 | automated_watcher | changed | 2026-08-05 11:00 CEST     | 13      | 0     | 1       | 13        | 28657 |
| 156 | automated_watcher | changed | 2026-08-15 16:00 CEST     | 2       | 0     | 11      | 2         | 28657 |
| 157 | automated_watcher | changed | 2026-08-15 18:00 CEST     | 3       | 1     | 0       | 2         | 28657 |
| 158 | automated_watcher | changed | 2026-08-16 03:00 CEST     | 4       | 1     | 0       | 3         | 28657 |
| 159 | automated_watcher | changed | 2026-08-16 09:00 CEST     | 5       | 1     | 0       | 4         | 28657 |
| 160 | automated_watcher | changed | 2026-08-16 10:00 CEST     | 6       | 1     | 0       | 5         | 28657 |
| 161 | automated_watcher | changed | 2026-08-16 13:00 CEST     | 7       | 1     | 0       | 6         | 28657 |
| 162 | automated_watcher | changed | 2026-08-17 03:00 CEST     | 8       | 1     | 0       | 7         | 28657 |
| 163 | automated_watcher | changed | 2026-08-18 03:00 CEST     | 9       | 1     | 0       | 8         | 28657 |

| #   | Phase             | Status  | Source Time           | Current | Added | Removed | Unchanged | Port  |
|-----|-------------------|---------|-----------------------|---------|-------|---------|-----------|-------|
| 164 | automated_watcher | changed | 2026-08-18 05:00 CEST | 10      | 1     | 0       | 9         | 28657 |
| 165 | automated_watcher | changed | 2026-08-19 16:00 CEST | 9       | 0     | 1       | 9         | 28657 |
| 166 | automated_watcher | changed | 2026-08-19 21:00 CEST | 10      | 1     | 0       | 9         | 28657 |
| 167 | automated_watcher | changed | 2026-08-20 18:00 CEST | 9       | 0     | 1       | 9         | 28657 |
| 168 | automated_watcher | changed | 2026-08-21 03:00 CEST | 8       | 0     | 1       | 8         | 28657 |
| 169 | automated_watcher | changed | 2026-08-21 06:00 CEST | 9       | 1     | 0       | 8         | 28657 |
| 170 | automated_watcher | changed | 2026-08-22 06:00 CEST | 8       | 0     | 1       | 8         | 28657 |
| 171 | automated_watcher | changed | 2026-08-22 11:00 CEST | 9       | 1     | 0       | 8         | 28657 |
| 172 | automated_watcher | changed | 2026-08-23 06:00 CEST | 8       | 0     | 1       | 8         | 28657 |
| 173 | automated_watcher | changed | 2026-08-23 12:00 CEST | 6       | 0     | 2       | 6         | 28657 |
| 174 | automated_watcher | changed | 2026-08-23 12:00 CEST | 8       | 2     | 0       | 6         | 28657 |
| 175 | automated_watcher | changed | 2026-08-25 03:00 CEST | 9       | 1     | 0       | 8         | 28657 |
| 176 | automated_watcher | changed | 2026-08-25 05:00 CEST | 10      | 1     | 0       | 9         | 28657 |
| 177 | automated_watcher | changed | 2026-08-26 21:00 CEST | 11      | 1     | 0       | 10        | 28657 |
| 178 | automated_watcher | changed | 2026-08-28 23:00 CEST | 10      | 0     | 1       | 10        | 28657 |
| 179 | automated_watcher | changed | 2026-08-29 12:00 CEST | 9       | 0     | 1       | 9         | 28657 |
| 180 | automated_watcher | changed | 2026-08-30 09:00 CEST | 10      | 1     | 0       | 9         | 28657 |
| 181 | automated_watcher | changed | 2026-08-30 15:00 CEST | 9       | 0     | 1       | 9         | 28657 |
| 182 | automated_watcher | changed | 2026-08-30 19:00 CEST | 10      | 1     | 0       | 9         | 28657 |
| 183 | automated_watcher | changed | 2026-09-03 13:00 CEST | 6       | 1     | 5       | 5         | 28657 |
| 184 | automated_watcher | changed | 2026-09-03 17:00 CEST | 7       | 2     | 1       | 5         | 28657 |
| 185 | automated_watcher | changed | 2026-09-04 20:00 CEST | 6       | 0     | 1       | 6         | 28657 |
| 186 | automated_watcher | changed | 2026-09-06 18:00 CEST | 7       | 1     | 0       | 6         | 28657 |
| 187 | automated_watcher | changed | 2026-09-06 20:00 CEST | 6       | 0     | 1       | 6         | 28657 |
| 188 | automated_watcher | changed | 2026-09-08 02:00 CEST | 6       | 1     | 1       | 5         | 28657 |
| 189 | automated_watcher | changed | 2026-09-08 07:00 CEST | 7       | 1     | 0       | 6         | 28657 |
| 190 | automated_watcher | changed | 2026-09-09 17:00 CEST | 6       | 0     | 1       | 6         | 28657 |
| 191 | automated_watcher | changed | 2026-09-09 20:00 CEST | 7       | 1     | 0       | 6         | 28657 |

| #   | Phase             | Status  | Source Time           | Current | Added | Removed | Unchanged | Port  |
|-----|-------------------|---------|-----------------------|---------|-------|---------|-----------|-------|
| 192 | automated_watcher | changed | 2026-09-11 08:00 CEST | 5       | 0     | 2       | 5         | 28657 |
| 193 | automated_watcher | changed | 2026-09-11 13:00 CEST | 6       | 1     | 0       | 5         | 28657 |
| 194 | automated_watcher | changed | 2026-09-12 08:00 CEST | 4       | 0     | 2       | 4         | 28657 |
| 195 | automated_watcher | changed | 2026-09-12 13:00 CEST | 6       | 2     | 0       | 4         | 28657 |
| 196 | automated_watcher | changed | 2026-09-13 08:00 CEST | 7       | 1     | 0       | 6         | 28657 |
| 197 | automated_watcher | changed | 2026-09-17 20:00 CEST | 8       | 1     | 0       | 7         | 28657 |
| 198 | automated_watcher | changed | 2026-09-19 05:00 CEST | 7       | 0     | 1       | 7         | 28657 |

## 7. Report-Use Notes

- Use the 7D report for short-range rotation and recent watcher movement.
- Use the 30D report for broader trend context once the 15-minute watcher has accumulated enough observations.
- Use the All Time report for full testnet observation history preserved by this watcher.
- For publication-quality interpretation, compare this Markdown output with dashboard charts, raw JSON outputs, and selected snapshots.